Levon Melkonyan

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Education

Universitat Pompeu Fabra Expected Aug 2028
Master of Science in Sound and Music Computing *Barcelona, Spain*

California Polytechnic State University, San Luis Obispo Sept 2024
Bachelor of Science in Computer Science *San Luis Obispo, CA*

Work Experience

VOICE (Visual Outputs for Inclusive Change and Environments) Jun 2024 - Present
Multimedia Software Engineer, *Part-time* *Remote*

- Designed and deployed an **ESP32/C++** interactive installation for the 2025 *Time Space Existence* exhibition in Venice, integrating motion sensing, motor control, and audio playback in a non-blocking state machine with watchdog recovery for 24/7 operation
- Developed an automated speech-processing pipeline using **PyTorch**, OpenAI Whisper, and FFmpeg to denoise, diarize, anonymize, transcribe, and caption recorded media
- Engineered a video anonymization pipeline using **OpenCV** and YOLOv8, with batched frame processing, temporal detection smoothing, selective face blurring, and automated re-encoding

CK Technologies, Inc. Mar 2025 - Apr 2026
Software Engineer, *Full-time* *Camarillo, CA*

- Architected an **STM32**-based overheat protection tool for 24/7 field operation, designing state machines, budgeting flash/RAM, defining safety and fail-safe behavior, and collaborating with electrical engineers on schematics, hardware interfaces, and test planning
- Implemented a hardware watchdog and interrupt-driven **C++** drivers for SD-card logging, RTC timestamps, and a TFT display with a custom library over SPI/I²C, debugging interfaces with a logic analyzer and oscilloscope
- Maintained **C++** test software and embedded firmware for aerospace cable and wire-harness systems, implementing hardware diagnostics and test routines for fault isolation and coverage
- Built an automated **Python** test suite with HTML/CSV report generation that simulates user I/O and verifies timing paths, reducing manual UI testing

Personal Projects — github.com/levon-m

MicroLoop: MIDI-Synced Live Looper & Sampler

- Built an **ARM Cortex-M7/C++17** performance looper and sampler with *beat-repeat*, *loop-freeze*, and *rhythmic gate* effects on live audio, including quantization to external MIDI clock, parameter presets, and an OLED menu system
- Implemented an SGTL5000 codec driver over I²C and custom audio I/O with click-free crossfades, double buffering, and a deterministic, zero-allocation DSP path
- Architected preemptive real-time firmware with an audio ISR and five control tasks communicating through non-blocking SPSC queues

BassMINT: Mount for Infrared Note Transcription

- Built an **ARM Cortex-M7/C++17** bass guitar bridge sensor that detects active string and estimates fret position in real time, streaming performance data over MIDI
- Combined IR-based string detection with YIN/autocorrelation pitch estimation from audio to infer fret position, using hand-position tracking to improve stability and reduce octave errors
- Integrated the embedded system with a cross-platform **JUCE** application featuring real-time fretboard visualization and performance statistics

Skills

Languages: C++, C, Python, Bash
Tools: CMake, GCC/Clang, GDB, JTAG, JUCE, PyTorch, OpenCV, Docker, Git, Oscilloscope, Logic Analyzer
Concepts: Real-time DSP, Multithreading, ISRs, RTOS, Embedded Systems, MIDI, Serial Protocols (UART, SPI, I²C, I²S)